DOCUMENTATION

PROJECT NAME:

"EPIDEMIC SPREAD OVER N WEEKS"

Team name: EPIDEMICS

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Course / Section: CSM

PROJECT THEME: predicting how a disease spreads through a population week by week, using a small table of probabilities (a matrix) and its powers

METHOD: A^n computed with the characteristic equation and the Cayley–Hamilton theorem

Page 1: Overview and Goal

THE REAL WORLD PROBLEM

- A disease spreads when healthy people meet sick people.
- Every week, some healthy people get sick, and some sick people get better.
- We want to know how many people are healthy and sick after 1 week, 10 weeks, and a full year (52 weeks).
- We also want to know if the disease dies out or settles at a steady level.

THE TWO STATES USED IN THE SAME VERSION

- S (Susceptible): healthy, can catch the disease.
- I (Infected): sick and can pass the disease on.
- Every person is in exactly one state each week.

THE TWO WEEKLY PROBABILITIES

- β (beta): chance that a healthy person becomes sick this week.

- γ (gamma): chance that a sick person recovers this week (and becomes healthy again).
- Example used throughout: β = 0.3 (30%) and γ = 0.4 (40%).

THREE QUESTIONS THIS PROJECT ANSWER

- What are S and I after each week from 1 to 52?
- Can we jump straight to week 52 without doing 52 separate steps? (Yes, using Cayley–Hamilton.)
- What is the long-run (steady) state?

WHY THIS MATTERS

- Every SIR-type COVID model repeats one weekly rule again and again.
- Repeating a rule n times is the same as raising a matrix to the power n.
- A fast way to compute A^n makes long forecasts cheap and exact.

Page 2: BASIC CONCEPTS IN SINGLE WORDS

BUILDING THE MATRIX A

- A healthy person stays healthy with probability 1 − β, or gets sick with probability β.
- A sick person stays sick with probability 1 − γ, or recovers with probability γ.
- Column 1 (from S): [1 − β, β]. Column 2 (from I): [γ, 1 − γ].
- With β = 0.3 and γ = 0.4: first column is [0.7, 0.3] and second column is [0.4, 0.6].
- So A = [0.7 0.4 ; 0.3 0.6] (first row 0.7, 0.4 and second row 0.3, 0.6).
- Check: each column adds to 1 (0.7 + 0.3 = 1 and 0.4 + 0.6 = 1), because everyone must go somewhere.

ONE WEEK OF SPREAD

- Write the population as x = (S, I).
- Next week: new S = 0.7 × S + 0.4 × I, and new I = 0.3 × S + 0.6 × I.
- In matrix form: x(next week) = A × x(this week).
- After n weeks: $x(n) = A^n \times x(0)$, where A^n means A multiplied by itself n times.

WHY WE NEED A SHORCUT

- Doing the weekly step 52 times is slow by hand and wasteful for big matrices.
- Cayley–Hamilton lets us write any power of A using only I and A (I is the identity matrix: 1s on the diagonal, 0s elsewhere).

- That means any week, even week 5,000,000, can be reached with a few simple numbers.

Page 3: THE MATHEMATICS WITH A WORKED EXAMPLE

Step 1: CHARACTERISTIC EQUATION

- For a 2×2 matrix the equation is $\lambda^2 - \tau\lambda + \Delta = 0$.
- Put in the numbers: $\lambda^2 - 1.3\lambda + 0.3 = 0$.
- Factor it: $(\lambda - 1)(\lambda - 0.3) = 0$, because $1 \times 0.3 = 0.3$ and $1 + 0.3 = 1.3$.
- So the eigenvalues are $\lambda = 1$ and $\lambda = 0.3$.
- General rule: the eigenvalues are always 1 and $(1 - \beta - \gamma)$.

Step 2: CALEY-HAMINTION THOEREM

- The theorem says: a matrix satisfies its own characteristic equation.
- Replace λ by A and the number 0.3 by $0.3 \times I$: $A^2 - 1.3A + 0.3I = 0$.
- Rearrange: $A^2 = 1.3A - 0.3I$.
- Check by direct multiplication: $A \times A$ = [0.61 0.52 ; 0.39 0.48].
- Check with the formula: $1.3A - 0.3I$ = [0.91 − 0.3 0.52 ; 0.39 0.78 − 0.3] = [0.61 0.52 ; 0.39 0.48]. Both agree.

Step 3: GETTING HIGH POWERS OF A $A^2 = 1.3A - 0.3I$

The big idea: the rule $A^2 = 1.3A - 0.3I$ lets us replace A^2 by something simpler. So whenever a power of A appears, we can shrink it back down to just "some number times A, plus some number times I".

Getting A^3. Multiply both sides of $A^2 = 1.3A - 0.3I$ by A:

- $A^3 = A \times A^2 = A \times (1.3A - 0.3I) = 1.3A^2 - 0.3A$.
- Now swap A^2 for $(1.3A - 0.3I)$: $A^3 = 1.3(1.3A - 0.3I) - 0.3A$.
- Expand: $A^3 = 1.69A - 0.39I - 0.3A$.
- Collect the A terms: **$A^3 = 1.39A - 0.39I$**.

Getting A^4. Do the same thing once more, starting from A^3:

- $A^4 = A \times A^3 = 1.39A^2 - 0.39A$.
- Swap A^2 again: $A^4 = 1.39(1.3A - 0.3I) - 0.39A = 1.807A - 0.417I - 0.39A$.
- So **$A^4 = 1.417A - 0.417I$**.

Every power has the same shape. Write $A^n = c_1 \cdot A + c_0 \cdot I$, where c_0 and c_1 are two plain numbers.

- $A^1 = 1 \cdot A + 0 \cdot I$, so $(c_0, c_1) = (0, 1)$.
- $A^2 = 1.3 \cdot A - 0.3 \cdot I$, so $(c_0, c_1) = (-0.3, 1.3)$.
- $A^3 = 1.39 \cdot A - 0.39 \cdot I$, so $(c_0, c_1) = (-0.39, 1.39)$.
- $A^4 = 1.417 \cdot A - 0.417 \cdot I$, so $(c_0, c_1) = (-0.417, 1.417)$.

The step-by-step rule. Multiplying A^n by A and swapping A^2 for $\tau A - \Delta I$ gives:

- new $c_0 = -\Delta \times$ (old c_1)
- new c_1 = (old c_0) + $\tau \times$ (old c_1)
- Example, from n = 3 to n = 4: new $c_0 = -0.3 \times 1.39 = -0.417$ and new $c_1 = -0.39 + 1.3 \times 1.39 = 1.417$. This matches the direct calculation above.

Step 4: A DIRECT FORMULA FOR ANY N (NO STEP BY STEP NEEDED)

The idea: the formula $A^n = c_0 \cdot I + c_1 \cdot A$ must also be true when A is replaced by either of its eigenvalues (1 and 0.3). That gives two simple equations, one for each eigenvalue.

- Using $\lambda = 1$: $1^n = c_0 + c_1 \cdot 1$, so $\mathbf{c_0 + c_1 = 1}$.
- Using $\lambda = 0.3$: $0.3^n = c_0 + c_1 \cdot 0.3$, so $\mathbf{c_0 + 0.3c_1 = 0.3^n}$.

Solving them. Subtract the second equation from the first:

- $(c_0 + c_1) - (c_0 + 0.3c_1) = 1 - 0.3^n$
- $0.7c_1 = 1 - 0.3^n$
- So $\mathbf{c_1(n) = (1 - 0.3^n) \div 0.7}$, and from the first equation $\mathbf{c_0(n) = 1 - c_1(n)}$.

Check n = 3: $0.3^3 = 0.027$, so $c_1 = 0.973 \div 0.7 = 1.39$ and $c_0 = 1 - 1.39 = -0.39$. This matches Step 4.

Week 52: 0.3^{52} is so tiny it is practically 0, so:

- $c_1 \approx 1 \div 0.7 = 1.4286$ and $c_0 \approx -0.4286$.
- $A^{52} \approx 1.4286 \cdot A - 0.4286 \cdot I$.
- Top-left: $1.4286 \times 0.7 - 0.4286 = 0.5714$. Top-right: $1.4286 \times 0.4 = 0.5714$.
- Bottom-left: $1.4286 \times 0.3 = 0.4286$. Bottom-right: $1.4286 \times 0.6 - 0.4286 = 0.4286$.
- So $A^{52} \approx$ [0.5714 0.5714 ; 0.4286 0.4286].

Step 5: USE A^3 TO FIND POPULATION AT WEEK 3

Build A^3 with $c_1 = 1.39$ and $c_0 = -0.39$:

- Top-left: $1.39 \times 0.7 - 0.39 = 0.973 - 0.39 = 0.583$.

- Top-right: 1.39 × 0.4 = 0.556.
- Bottom-left: 1.39 × 0.3 = 0.417.
- Bottom-right: 1.39 × 0.6 − 0.39 = 0.834 − 0.39 = 0.444.
- So A^3 = [0.583 0.556 ; 0.417 0.444].

Apply it to the starting population x(0) = (0.99, 0.01), meaning 99% healthy and 1% sick:

- Week 3 healthy = 0.583 × 0.99 + 0.556 × 0.01 = 0.5772 + 0.0056 ≈ 0.5827.
- Week 3 sick = 0.417 × 0.99 + 0.444 × 0.01 = 0.4128 + 0.0044 ≈ 0.4173.
- Check: 0.5827 + 0.4173 = 1, as it must be.

Apply A^{52} the same way to see the long run:

- Week 52 healthy = 0.5714 × 0.99 + 0.5714 × 0.01 = 0.5714.
- Week 52 sick = 0.4286 × 0.99 + 0.4286 × 0.01 = 0.4286.
- So after a year the population is about 57.14% healthy and 42.86% sick, reached with one formula instead of 52 weekly steps.

Page:4 WEEK BY WEEK EVOLUTION AND STEADY STATE

DOING THE WEEK ONE AT A TIME

- Week 0: S = 0.99 and I = 0.01.
- Week 1: S = 0.7 × 0.99 + 0.4 × 0.01 = 0.697 and I = 0.3 × 0.99 + 0.6 × 0.01 = 0.303.
- Week 2: S = 0.7 × 0.697 + 0.4 × 0.303 = 0.6091 and I = 0.3909.

FINDING THE STEADY STATE

- Steady state means the numbers stop changing: A × π = π.
- People moving from healthy to sick equal people moving from sick to healthy.
- That balance is β × S = γ × I, so 0.3 × S = 0.4 × I.
- With S + I = 1: S = 0.4 ÷ 0.7 = 0.5714 and I = 0.3 ÷ 0.7 = 0.4286.
- General rule: S = γ ÷ (β + γ) and I = β ÷ (β + γ).
- Check: 0.7 × 0.5714 + 0.4 × 0.4286 = 0.4 + 0.1714 = 0.5714. The healthy fraction stays the same.

WHY CALEY-HAMINTION IS FASTER

- Direct method: multiply A by itself n times, so the work grows with n.
- Cayley–Hamilton: update only two numbers (c_0, c_1) each step.

- With repeated squaring, only about $\log_2(n)$ steps are needed. For n = 52 that is around 6 squarings.
- The same idea works for bigger matrices (3×3, 4×4, SEIR models).

Page 5: CHATS VERIFICATION AND CONCLUSION

Better charts

- A clear title that names the method and the team (EPIDEMICS).
- Axis labels: "Week" on the horizontal axis and "Population fraction" on the vertical axis.
- A legend that separates Susceptible and Infected lines.
- Dashed horizontal lines showing the steady-state values.

VERIFICATION TESTS

- Compare Cayley–Hamilton A^n with NumPy's matrix power for n = 1, 2, 3, 10, 52, 200.
- Check that every column of A adds up to 1.
- Check that $A^2 = \tau A - \Delta I$ holds for the example.
- Check that each week's S + I equals 1.
- All differences should be near 10^{-15} (essentially rounding error).

CONCLUSION

- A year of epidemic spread is one matrix power, A^{52}.
- The characteristic equation $\lambda^2 - 1.3\lambda + 0.3 = 0$ gives eigenvalues 1 and 0.3 in our example.
- Cayley–Hamilton turns A^{52} into $1.4286 \cdot A - 0.4286 \cdot I$, with no repeated multiplication needed.
- The result is the steady state: about 57.14% healthy and 42.86% sick.
- Every step was checked by simple arithmetic and by direct multiplication.

THANK YOU ,EVERYONE